

So Let's Get Started

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CS703-152W: Applied Data Science Project

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Phase 1: Data Science Project Proposals

Project 1: Housing Amenities Impact on Apartment Listing Prices in the Southern USA

Business Issue and Overview

Real estate investors and property managers in the Southern USA lack quantitative evidence on which amenities drive rental pricing for apartments. Understanding amenity importance enables strategic property improvements and competitive pricing strategies. This project analyzes apartment listings and determines which amenities (parking, laundry, pet policies etc.) most significantly influence rental prices in Southern apartment markets: Austin TX, Dallas TX, Miami FL, Atlanta GA, Nashville TN, and Charlotte NC.

Business Questions

- Which amenities have strongest impact on rental pricing?
- How does amenity importance vary across geographic regions?
- Which amenity upgrades yield the highest price premium relative to cost?
- Do certain amenity bundles (e.g., parking & laundry) have compounding effects on price?

Data Needs

Major entities: Apartment rental listings from Southern markets (Austin, Dallas, Miami, Atlanta, Nashville, Charlotte) including price, location, bedrooms, bathrooms, square footage, amenities (parking, laundry, pet policies, etc), listing metadata.

Reason

Addresses urgent business problem in high-growth Southern real estate markets. Understanding which amenities justify investment is directly actionable for property managers operating in Southern markets. Regional focus ensures recommendations are relevant to Southern investors rather than generic national insights.

Project Selection

After reviewing both proposals and considering available data sources, I have selected **Project 1: Housing Amenities Impact on Apartment Listing Prices in the Southern USA** for this semester's capstone project.